

Lab 1 — Requirements Engineering & UML Use-Case Modelling

Problem Statement #24 — Public Bus Live Tracking & Crowding Estimator

SRN	PES1UG24CS392
Primary Domain	Smart Cities, Transport & Logistics
Target Actors	Commuter, Fleet Controller
Repository	github.com/LOKESH-ADITHYA/SE-Labs-PES1UG24CS392

Problem Context

A municipal transit intelligence platform ingesting GPS feeds from city buses, estimating arrival times (ETA) at downstream stops, and computing passenger crowding levels from ticketing sensors.

Contents

1	Requirements Table — 5 functional (FR-001...FR-005) and 2 non-functional (NFR-001, NFR-002) requirements with Req ID, Type, Description, Priority, Acceptance Criteria and Rationale	p. 2–3
2	UML Use-Case Diagram — 5 actors, 10 use cases (UC-01...UC-10), 4 «include» and 2 «extend» relationships	p. 4
3	Use-Case Flow Specification — UC-02 View Live Bus Location & ETA: preconditions, postconditions, main success scenario and one alternate flow	p. 5

Each deliverable is also committed to the repository individually, in the formats named in the lab handout: the requirements table as Word, the use-case diagram as PDF, and the flow specification as both Word and PDF.

Lab 1 - Requirements Table

Problem Statement #24 - Public Bus Live Tracking & Crowding Estimator

Domain: Smart Cities, Transport & Logistics | Actors: Commuter, Fleet Controller | SRN: PES1UG24CS392

1. Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest bus GPS coordinates every 5 seconds and recalculate arrival ETAs for all upcoming route stops.	High	Pass: ETA for each downstream stop is updated within one 5-second cycle, computed from current traffic speed and remaining route distance. Fail: ETA increases while the bus approaches the stop, or the ETA shown is based on telemetry older than 15 seconds.	ETA accuracy is the core value of the platform; every commuter-facing feature depends on this figure, and a stale feed makes predictions drift beyond usefulness.
FR-002	Functional	The system shall compute an occupancy figure for each active bus from onboard ticketing / passenger-counting sensor events and classify it into a crowding band: Seats Available (< 50% of rated capacity), Standing Room (50-85%), Full (> 85%).	High	Pass: For a given sequence of boarding and alighting events the occupancy is computed correctly, the band matches the stated thresholds, and the band is refreshed on every stop departure. Fail: Occupancy goes negative, exceeds rated capacity without raising Full, or the band is not updated after a stop event.	Crowding is the second capability named in the problem statement. Commuters need it to decide whether to board or wait; controllers need it to detect overloaded services.
FR-003	Functional	The system shall allow a Commuter to search by route number or by stop and shall display, for every approaching bus, its live position on a map, its ETA at that stop, and its current crowding band.	High	Pass: A search on a valid route or stop returns all approaching buses within 3 seconds, each entry carrying position, ETA and crowding band. Fail: A valid route or stop returns no result, or a listed bus is missing its ETA or crowding band.	The primary commuter-facing interaction and the entry point to every other commuter feature; without it the computed data never reaches a user.
FR-004	Functional	The system shall allow a Commuter to subscribe to an arrival alert for a chosen bus at a chosen stop and shall push a notification when the predicted ETA falls below the commuter's chosen lead time (default 5 minutes).	Medium	Pass: Exactly one notification is delivered within 30 seconds of the ETA crossing the configured threshold, naming the bus, the stop and the remaining minutes. Fail: No notification is delivered, duplicates are sent for one subscription, or a notification arrives after the bus has departed.	Removes the need to keep the app open and watch the map - the main convenience the platform offers to a commuter waiting at a stop.
FR-005	Functional	The system shall present the Fleet Controller with a dashboard of all active buses, automatically flag any bus delayed beyond a configurable threshold (default 10 minutes) or holding a Full crowding band, and allow the controller to dispatch a relief bus onto the affected route.	High	Pass: A bus breaching either threshold is shown as flagged within one dashboard refresh cycle; dispatching a relief bus creates an assignment record and the relief bus appears as active on that route. Fail: A breaching bus is not flagged, or a dispatch action completes without creating an assignment record.	Turns passive monitoring into operational control. The platform is "transit intelligence" only if the operator can act on what it detects.

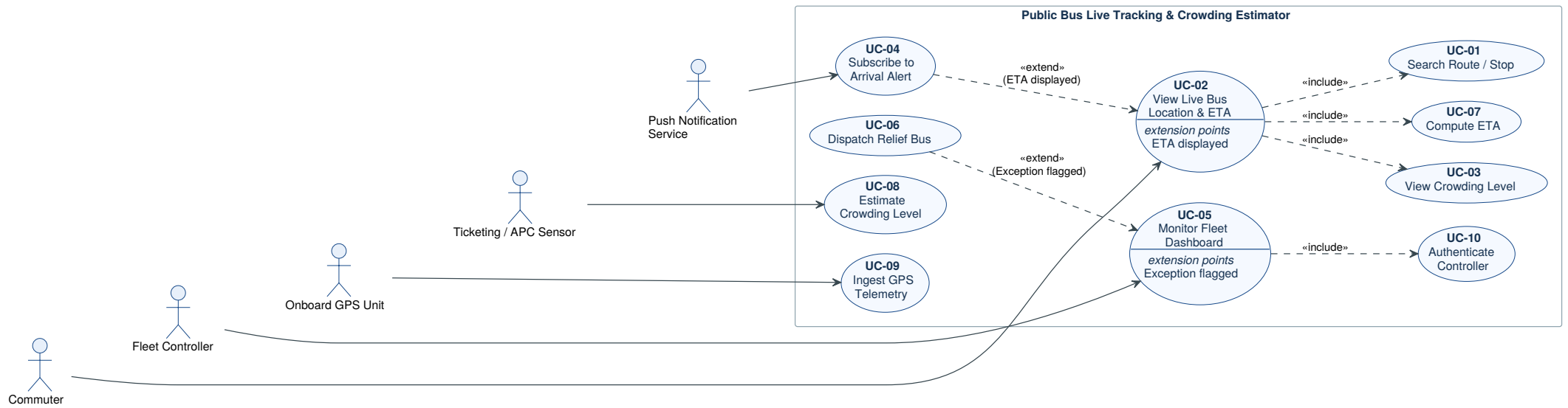
2. Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional - Performance & Scalability	The tracking engine shall handle telemetry streams from up to 1,000 active transit vehicles simultaneously and complete a full ETA recomputation cycle across the fleet with a 95th-percentile latency under 2 seconds.	High	Pass: A 30-minute benchmark at 1,000 simulated vehicles sustains p95 ETA-refresh latency below 2 seconds with packet loss under 0.1%. Fail: p95 latency reaches 2 seconds or more, or packet loss exceeds 0.1% under the same simulated peak load.	A mid-size city fleet is on the order of a thousand vehicles. If the pipeline cannot keep pace at peak hour, ETAs go stale and FR-001 and FR-003 fail in practice even though their logic is correct.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Nonfunctional - Security & Privacy	All client-server and telemetry traffic shall use TLS 1.3; Fleet Controller functions shall be gated behind role-based access control; commuter location data used for nearest-stop lookup shall be purged within 24 hours and never linked to a persistent identifier.	High	Pass: A TLS scan finds no plaintext or down-level endpoint; a Commuter-role token is rejected with HTTP 403 on every controller endpoint; an audit log shows the purge job removing all location records older than 24 hours. Fail: Any endpoint accepts plaintext, a Commuter-role token reaches a controller function, or location records survive beyond 24 hours.	Dispatch controls are safety-relevant and must never be publicly actionable, and commuter location traces are personal data subject to India's DPDP Act, 2023.

3. Traceability Summary

Requirement	Realised by Use Case(s)
FR-001	UC-09 Ingest GPS Telemetry, UC-07 Compute ETA
FR-002	UC-03 View Crowding Level, UC-08 Estimate Crowding Level
FR-003	UC-01 Search Route / Stop, UC-02 View Live Bus Location & ETA
FR-004	UC-04 Subscribe to Arrival Alert
FR-005	UC-05 Monitor Fleet Dashboard, UC-06 Dispatch Relief Bus
NFR-001	Constrains UC-09 Ingest GPS Telemetry and UC-07 Compute ETA
NFR-002	Constrains UC-10 Authenticate Controller and all data paths



Use-Case Flow Specification - UC-02 View Live Bus Location & ETA

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Use Case	UC-02 - View Live Bus Location & ETA
Primary Actor	Commuter
Secondary Actors	Onboard GPS Unit, Ticketing / APC Sensor
Related Requirements	FR-001, FR-002, FR-003 (constrained by NFR-001, NFR-002)
Includes	UC-01 Search Route / Stop, UC-07 Compute ETA, UC-03 View Crowding Level
Extended by	UC-04 Subscribe to Arrival Alert, at extension point "ETA displayed"
Trigger	The Commuter requests live arrival information for a route or stop.

Preconditions

- The Commuter has the application open and the device has network connectivity.
- At least one bus is in active service on the selected route, and its Onboard GPS Unit has reported a position within the last 15 seconds.
- Route stop sequence, timetable and rated vehicle capacity are loaded in reference data.

Postconditions

- Success:** the Commuter is shown each approaching bus with its live map position, ETA at the selected stop and crowding band, refreshing at the 5-second telemetry cadence for as long as the view stays open.
- On failure:** the Commuter is told explicitly that live data is unavailable and is shown the scheduled timetable instead; no ETA is presented as if it were live.

Main Success Scenario

#	Actor	Action
1	Commuter	Requests arrival information for a route or stop.
2	System	<<include>> UC-01 Search Route / Stop - validates the input against reference data and retrieves every in-service bus whose remaining path includes the selected stop.
3	System	<<include>> UC-07 Compute ETA - for each such bus, computes arrival time at the stop from the latest GPS fix, remaining route distance and current traffic speed.
4	System	<<include>> UC-03 View Crowding Level - for each such bus, retrieves the crowding band computed from ticketing / APC sensor counts and prepares it for display.
5	System	Renders each bus at its live position on the map, listed with ETA in minutes and a colour-coded crowding badge, sorted by soonest arrival. (Extension point: ETA displayed.)
6	Commuter	Reviews the list and selects a bus to see its stop-by-stop progress.
7	System	Refreshes position, ETA and crowding band every 5 seconds as new telemetry arrives, until the Commuter leaves the view.

Alternate Flow - 5a. GPS telemetry stale or unavailable (branches at step 3)

5a.1	System detects that the latest GPS fix for a bus is older than 15 seconds (dropped signal, tunnel, or onboard unit fault).
5a.2	System suppresses the live ETA for that bus rather than extrapolating from the stale fix.
5a.3	System substitutes the scheduled arrival time from the timetable, marks the entry "Scheduled - live tracking unavailable", and greys out the bus icon at its last known position with the age of that fix shown.
5a.4	System raises a telemetry gap event on the Fleet Controller dashboard for investigation (feeds FR-005).
5a.5	When a fresh GPS fix arrives, the entry reverts to a live ETA on the next refresh cycle and the flow rejoins the main scenario at step 5.

Outcome: the Commuter still receives usable arrival guidance and is never misled into treating a scheduled time as a live prediction.